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SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)
(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC
Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

A PROJECT REPORT

ON

"MULTI-MODAL GRAPHRAG: A CONTROLLED STUDY OF CONTEXT MANAGEMENT, CROSS-MODAL FUSION AND RETRIEVAL CONTROL"

A project report submitted in partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering

BACHELOR OF TECHNOLOGY COMPUTER SCIENCE AND ENGINEERING

Submitted By
Aaditya Bhure (PRN: [PRN])

UNDER THE GUIDANCE OF

[Guide Name]
[Designation]

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
AY 2025-26

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Project work entitled "**Multi-Modal GraphRAG: A Controlled Study of Context Management, Cross-Modal Fusion and Retrieval Control over Text, Graph and Structured Knowledge**" is carried out by **Aaditya Bhure**, in partial fulfillment for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering**, Symbiosis International (Deemed University), Pune during the academic year 2025-2026.

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DECLARATION

I hereby declare that the project titled "**Multi-Modal GraphRAG: A Controlled Study of Context Management, Cross-Modal Fusion and Retrieval Control over Text, Graph and Structured Knowledge**" submitted to Symbiosis Institute of Technology, Constituent of Symbiosis International (Deemed University) Pune for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a result of original research carried out by me. I understand that my report may be made electronically available to the public. It is further declared that the project report or any part thereof has not been previously submitted to any University or Institute for the award of any degree or diploma.

Name of the Student	: Aaditya Bhure
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Degree	: Bachelor of Technology in CSE
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_____

(Signature of the Student)

Date:

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ABSTRACT

Retrieval-augmented generation (RAG) lets large language models (LLMs) answer from external evidence, and real collections increasingly hold the same entities as unstructured text, knowledge graphs and structured records. Systems that combine these sources change how sources are queried, how evidence is fused and how the context is managed all at once, so it remains unclear which design decision matters.

This project builds a reproducible platform for a controlled, one-variable-at-a-time study of multi-modal GraphRAG. Three strictly isolated views (text, graph, structured) of the STaRK-MAG scholarly corpus (700,244 papers, 1.87 million entities, 19.9 million relations) are joined only by a global identifier, and a benchmark is constructed whose answers are computed before questions are worded and whose multi-source questions provably require every source. With frozen retrievers and one locally served LLM, fifteen strategies are compared (five each for context management, cross-modal fusion and retrieval control, including adaptations of nine published methods), and the best are integrated. Every model call is stored and replay-verified, differences are tested with family-clustered paired statistics and Holm correction, and an interactive application replays every recorded run.

No literature-inspired method improved answer quality with statistical support. A deterministic rule router selected sources significantly better than querying every source (source-selection F1 0.876 vs. 0.700, Holm-adjusted $p = 0.014$) at the lowest cost, whereas learned and agentic controllers spent up to 6,096 more tokens per question for identical answers. The winners of the three components combine into the integrated system selected by a rule fixed in advance. Answer quality is bounded upstream: on average at most 34% of the correct papers reached the model's context.

Keywords: Retrieval-augmented generation, GraphRAG, knowledge graphs, heterogeneous data fusion, context management, adaptive retrieval, LLM agents, reproducibility, statistical testing

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LIST OF ABBREVIATIONS

Abbreviation	Full form
API	Application programming interface
AR	Adaptive retrieval (retrieval-control policy)
CF	Cross-modal fusion (strategy)
CI	Confidence interval
CM	Context management (strategy)
COCOMO	Constructive Cost Model
CRAG	Corrective retrieval-augmented generation
CSR	Compressed sparse row
DFD	Data-flow diagram
EM	Exact match
FAISS	Facebook AI Similarity Search
FR / NFR	Functional / non-functional requirement
GC	Gold coverage
GPU	Graphics processing unit
INT-1	The integrated system of this project
JSON	JavaScript Object Notation
KAG	Knowledge-augmented generation
KG	Knowledge graph
KLOC	Thousands of lines of code
LLM	Large language model
MAG	Microsoft Academic Graph
MRR	Mean reciprocal rank
PCST	Prize-collecting Steiner tree
QAFD	Query-aware flow diffusion
RAG	Retrieval-augmented generation
RG / RQ	Research gap / research question
SAG	SQL-retrieval augmented generation
SF1	Source-selection F1
SQL	Structured Query Language
SRS	Software requirements specification
WBS	Work breakdown structure
WR	Wasted retrieval

Chapter 1
Introduction

1.1 Introduction

Large language models (LLMs) answer questions fluently but cannot consult facts they were never shown, and they can state wrong facts with confidence. Retrieval-augmented generation (RAG) retrieves evidence from an external collection and asks the model to answer only from that evidence [1], [2]. Real collections, however, describe the same entities in several forms. A scholarly corpus contains what each paper *says* (title and abstract), how papers and people are *connected* (authorship, affiliation, citation, topic) and facts *recorded* about each paper (year, citation count, rank, venue). A question such as "papers whose abstract states *X*, written by an author affiliated with institution *Y*, published in a journal with a rank of at most *Z*" needs all three representations: dense text retrieval, graph retrieval (GraphRAG) and structured retrieval.

A system that answers such questions must decide **which sources to query, in what order and when to stop** (retrieval control), **how to combine evidence of very different shapes** (cross-modal fusion) and **which evidence survives a limited context budget** (context management). Each decision has attracted elaborate methods [3]–[11], but each is proposed inside its own pipeline and evaluated on its own corpus, retrievers, model and budget. This project started as a small demonstration comparing traditional RAG with basic GraphRAG over twelve papers and was rebuilt into a research platform that compares these decisions under strictly controlled conditions.

1.2 Problem Statement

Because integrated systems change several design decisions at once, a reported gain cannot be attributed to any single component. Formally, let a corpus offer a text, a graph and a structured view of the same papers, joined only by a global identifier. Each question q has a verified answer set $G(q)$ of paper identifiers, possibly empty. A system queries some views, builds a context within a fixed budget and asks a fixed LLM for an answer set $A(q)$ containing only identifiers present in the context. For each component in turn, with all others fixed, the task is to find the strategy that makes $A(q)$ match $G(q)$ most closely at low cost, to accept differences only when a paired, correlation-aware test supports them, and finally to measure the combination of the chosen strategies (Table 1.1). Figure 1.1

shows why such questions are hard: each source alone admits between 111 and 130,859 papers, while exactly one paper satisfies all conditions.

Table 1.1: Research questions of the project

ID	Component	Research question
RQ1	Context management	How should retrieved graph evidence be selected, ordered and trimmed to survive a fixed context budget?
RQ2	Cross-modal fusion	How should text, graph and structured evidence be combined into one context?
RQ3	Retrieval control	Can a policy decide which sources to query, in what order and when to stop, without losing answers?
RQ4	Integration	What is the combined effect of the best choice for each component?

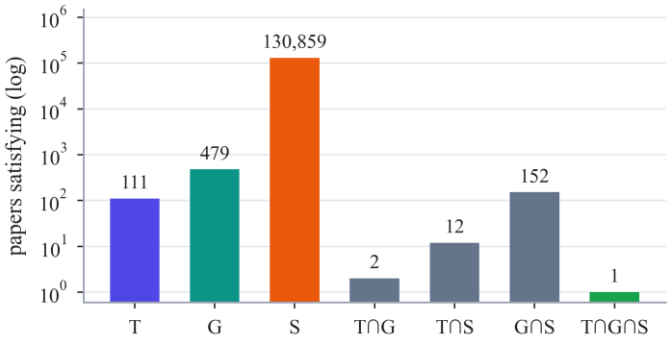


Figure 1.1: Candidate papers satisfying the conditions of one three-source question, per source and intersection (log scale); only the paper in the triple intersection is correct

1.3 Scope of Research

The project covers the full path from a raw public dataset to a statistically analysed comparison and an interactive presentation. **In scope:** the official STaRK-MAG corpus; three isolated views; a verified benchmark with noisy, conflicting and unanswerable variants; fifteen strategies and one integrated system; one locally served LLM; quality and cost metrics with paired statistics; a replay-verified protocol and an interactive application. **Out of scope:** other corpora or languages, LLM-extracted graphs (which would entangle the sources), training new retrievers or fine-tuning the LLM, comparisons across LLMs, and the locked test split, which awaits human review.

1.4 Research Hypotheses

Each research question gives a hypothesis reflecting the expectation from the literature that more elaborate mechanisms help (Table 1.2); the null hypothesis in each case is that they make no difference. Outcomes are reported in Section 5.8.

Table 1.2: Research hypotheses

ID	Hypothesis	Tested by
H1	A graph-aware context strategy yields higher answer F1 than fixed top-K selection under the same budget	RQ1, 8 families
H2	Entity-aligned, selective or late fusion yields higher answer F1 than labelled source sections	RQ2, 8 families
H3	Learned, corrective or agentic control selects sources better, or more cheaply, than a rule router	RQ3, 14 families
H4	Combining the best components selects sources better than querying every source, without losing answers	RQ4, 14 families

1.5 Objectives

1. To acquire and audit the official STaRK-MAG release and convert it into three isolated views joined only by the global identifier.
2. To construct a benchmark whose answers are computed before wording and whose multi-source questions provably require every source.
3. To implement simple retrievers, a grounded answer generator and fifteen strategies, including adaptations of nine published methods, behind one frozen interface.
4. To execute all studies under a protocol that freezes every input by checksum and verifies every result by replay.
5. To evaluate quality and cost with family-clustered paired statistics, Pareto sets and an overall ranking, and to integrate the winners.
6. To build an interactive application that presents every result from a checksummed data bundle.

1.6 Organization of the Report

Chapter 2 reviews related work and derives the research gaps. Chapter 3 specifies tools, hardware, work breakdown, requirements and cost. Chapter 4 describes the architecture, modules, algorithms, experimental design, execution protocol and testing. Chapter 5 reports and discusses the results, and Chapter 6 concludes with the future scope.

Chapter 2
Literature Review

2.1 Background

Retrieval-augmented generation. RAG conditions a generator on retrieved passages [1]; surveys organise its many variants [2]. Models use long contexts unevenly [12], and selecting or compressing evidence can cut cost without losing answers [13]; active and self-reflective variants decide during generation what to retrieve [14], [15]. Dense retrieval embeds questions and passages with sentence encoders [16], [17] and searches them with vector indexes such as FAISS [18].

Graph-based RAG. GraphRAG builds an entity graph with an LLM and summarises its communities [19]; G-Retriever selects a subgraph as a prize-collecting Steiner tree [3], [20], [21]; KG²RAG expands semantic seeds through a knowledge graph and organises them into reranked paragraphs [4]; QAFD-RAG diffuses relevance along query-aware edge weights [5], [22]; Think-on-Graph and HippoRAG explore the graph by LLM beam search and personalised propagation [23], [24]; a survey reviews the area [25].

Fusion of heterogeneous knowledge. UniK-QA linearises tables and triples [26], StructGPT reasons over structured interfaces [27], HybridRAG concatenates vector and graph retrieval [28], KAG aligns graph entities with text chunks [6], SAG selects structural evidence through SQL [7], MegaRAG reasons per modality before fusing [8], and query-driven multimodal GraphRAG builds local graphs on demand [29].

Adaptive and agentic retrieval control. Adaptive-RAG routes questions with a small T5 classifier [9], [30]; CRAG triggers corrections from a retrieval evaluator [10]; interleaved and reasoning-acting agents choose the next retrieval step [31], [32]; AGENT-G adds a retriever bank with validator and commentor critics [11]; and agent-orchestrated RAG adds decomposition and reflection [33].

Benchmarks and statistics. STaRK evaluates retrieval over semi-structured knowledge bases, including the Microsoft Academic Graph used here [34], [35]. Methodological work stresses significance testing on dependent samples [36]; clustered resampling, permutation tests, the bootstrap and Holm correction address this [37]–[40].

Table 2.1 summarises the methodology, datasets, main results and role in this project of the method papers that the project draws on.

Table 2.1: Literature synthesis of the method papers and their role in this project

Work	Methodology	Key datasets	Main results	Role here
KG²RAG [4]	Semantic seeds, KG-guided expansion and organisation	HotpotQA and variants	Better response and retrieval quality	CM-01
G-Retriever [3]	PCST subgraph, GNN encoder, soft prompt	GraphQA (ExplaGraphs, SceneGraphs, WebQSP)	Scales beyond the context window	CM-02
QAFD-RAG [5]	Training-free query-aware flow diffusion	UltraDomain, HotpotQA, MuSiQue, Spider 2.0	Gains over graph RAG baselines	CM-03
KAG [6]	Mutual indexing, logical-form reasoning	HotpotQA, 2WikiMultiHopQA, MuSiQue	+19.6% / +33.5% relative F1	CF-01
SAG [7]	Event–entity SQL index, dynamic hyperedges	HotpotQA, 2WikiMultiHopQA, MuSiQue	80.36% Recall@5 on MuSiQue	CF-02
MegaRAG [8]	Multimodal KG, two-stage generation	UltraDomain, multimodal documents	Beats GraphRAG, LightRAG	CF-03
Adaptive-RAG [9]	T5 classifier routes retrieval depth	SQuAD, NQ, TriviaQA, MuSiQue, HotpotQA	Better accuracy–efficiency balance	AR-01
CRAG [10]	Evaluator triggers corrective retrieval	PopQA, Biography, PubHealth, ARC	Gains for RAG approaches	AR-02
AGENT-G [11]	Agent, retriever bank, critic	STaRK (MAG, PRIME), CRAG	+47% Hit@1 on STaRK-MAG	AR-03
Think-on-Graph [23]	LLM beam search over a KG	CWQ, WebQSP, GrailQA	Traceable multi-hop reasoning	design
QD-MM GraphRAG [29]	Query-driven local graph construction	MultimodalQA, WebQA	Best unsupervised method	design
STaRK [34]	Semi-structured retrieval benchmark	STaRK-MAG, Amazon, PRIME	Benchmark for LLM retrieval	data

2.2 Summary of Literature Review and Research Gap

The literature offers many candidate answers to RQ1–RQ3 but no common ground on which to compare them: effects are not attributable to a single component, sources are often entangled (a graph extracted from the same text that is also retrieved), multi-source questions are rarely verified to need every source, and cost and statistical uncertainty are under-reported [36]. Table 2.2 lists the resulting research gaps and Figure 2.1 connects them to the research questions, strategies and findings.

Table 2.2: Research gaps derived from the literature review

Gap	Description	Addressed by
RG1	No controlled comparison of graph-context strategies under one budget with identical evidence	RQ1
RG2	No controlled comparison of fusion strategies on byte-identical multi-source evidence	RQ2
RG3	No controlled comparison of source-selection policies with the same retrievers	RQ3
RG4	Unknown combined effect of empirically chosen components	RQ4

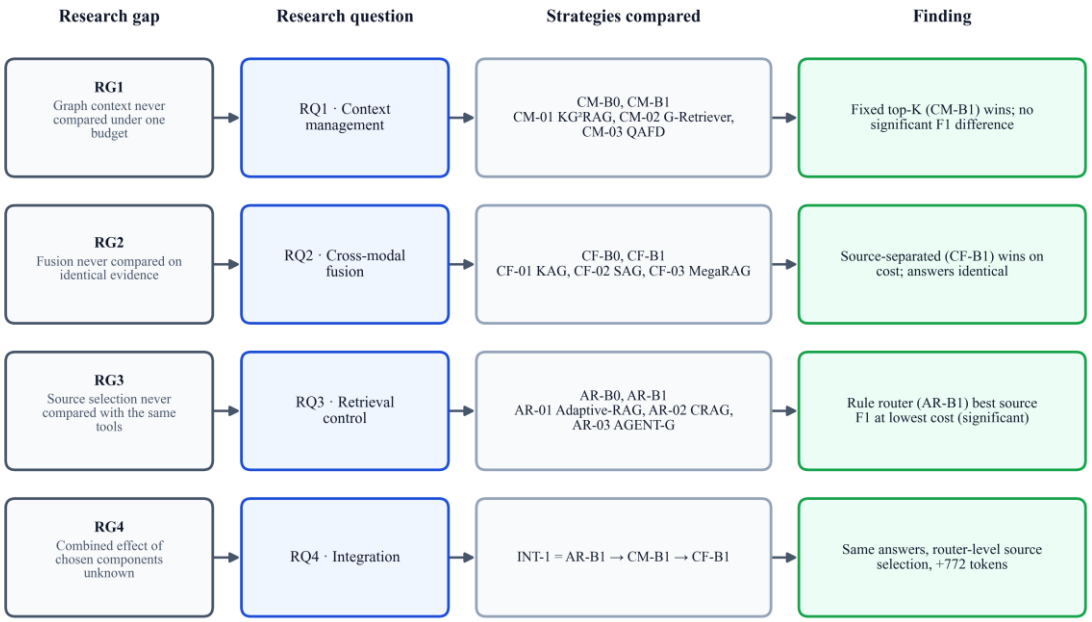


Figure 2.1: Research framework: gaps, research questions, strategies compared and findings

Chapter 3
Software Requirements Specification

3.1 Software Tool Platform, Tools and Frameworks Used

The platform is written in Python and uses only open-source software with pinned versions (Table 3.1).

Table 3.1: Software tools and frameworks

Tool / framework	Version	Purpose
Python	3.12.14	Research code, experiments, analysis
Ollama, gpt-oss:20b	0.40.5; 20.9 B, MXFP4	Local serving of the shared answer model [41], [42]
PyTorch, Transformers	2.14.0, 5.17.0	T5 classifier, cross-encoders, rerankers
sentence-transformers, ONNX Runtime	6.1.0, 1.23.2	MiniLM embeddings exported to ONNX
FAISS, DuckDB	1.15.1, 1.5.6	Exact vector index; read-only structured retrieval
NumPy, SciPy, scikit-learn	2.5.3, 1.18.1, 1.9.1	CSR graph, statistics, Platt scaling
NetworkX, pcst_fast	3.6.1, 1.0.10	Graph algorithms; Steiner tree solver
Pydantic, pytest	2.13.5, 9.1.1	Typed contracts; 327 automated tests
Streamlit, Altair, pandas	1.64.0, 6.3.0, 3.0.6	Interactive application and charts
Git, Matplotlib	—, 3.10.7	Version control; figures

3.2 Hardware Tools

All work ran on one laptop; the LLM was served locally, so no data left the machine: **Intel Core i7-13650HX** processor, **16 GB RAM**, **NVIDIA GeForce RTX 4060 Laptop GPU (8 GB)**, **Windows 11**. The frozen data occupy 7.17 GB in 695 checksummed files, and a cooling guard paused requests while the GPU was above 80 °C (never counted as latency).

3.3 Work Breakdown Structure

The work was organised into seven work packages (Figure 3.1) executed in strictly ordered phases between 4 and 18 October 2026 (Figure 3.2). Every phase ended with a review report, a checkpoint commit and an explicit approval; earlier results were preserved rather than overwritten.

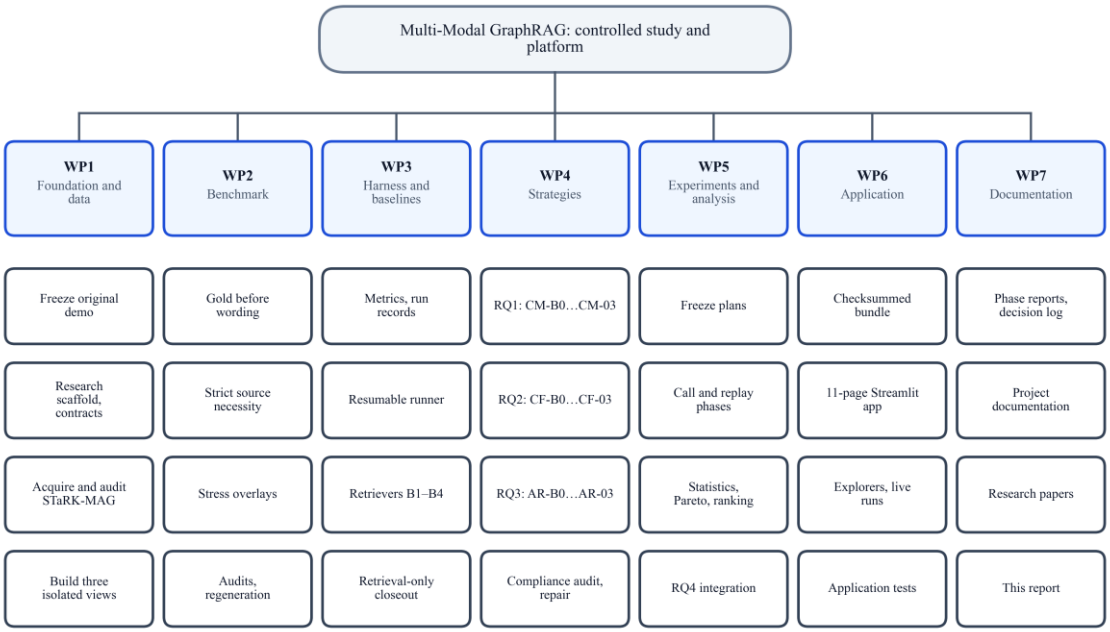


Figure 3.1: Work breakdown structure: seven work packages and their main tasks

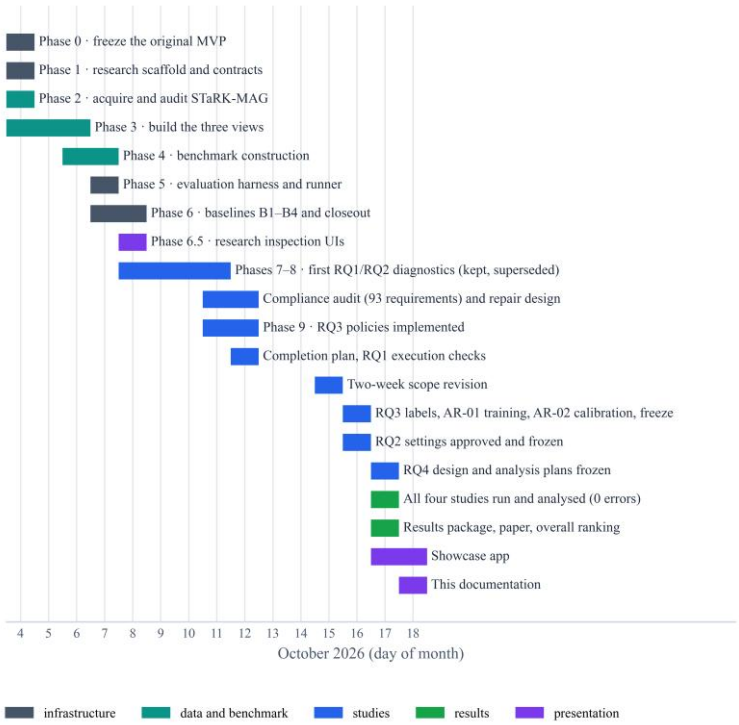


Figure 3.2: Project timeline, 4–18 October 2026

3.4 Functional Requirements

Table 3.2 lists the functional requirements and Figure 3.3 the use cases of the two kinds of user.

Table 3.2: Functional requirements

ID	Requirement	Layer
FR1	Download the pinned STaRK-MAG release and audit every record without modifying it	Data
FR2	Build three views sharing only the global identifier and audit their isolation	Data
FR3	Compute gold answers before wording, verify source necessity, add stress variants	Benchmark
FR4	Provide one retriever per source (B1–B3) and a fixed multi-source baseline (B4)	Retrieval
FR5	Provide fifteen strategies behind one frozen interface	Components
FR6	Answer with one grounded prompt; accept only identifiers present in the context	Generation
FR7	Freeze studies by checksum; store every model request; rebuild results by replay	Execution
FR8	Compute metrics, paired family-clustered tests, Pareto sets and an overall ranking	Analysis
FR9	Export a checksummed bundle; present, explore and replay every recorded run	Application

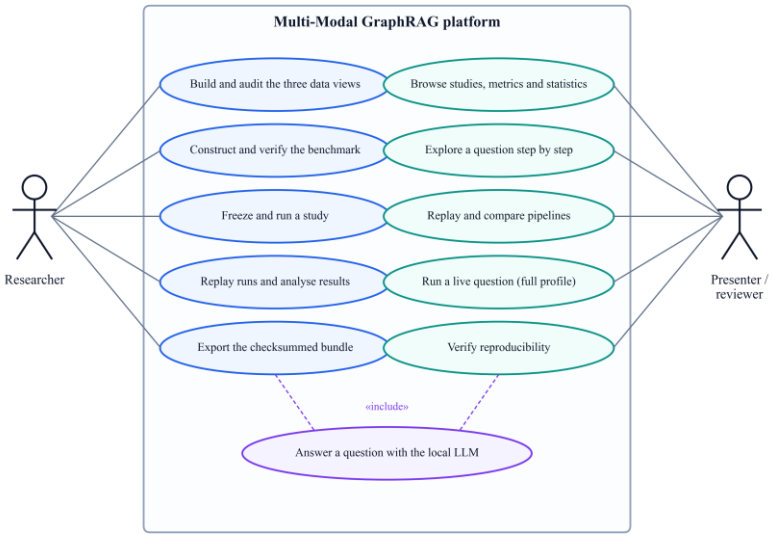


Figure 3.3: Use-case diagram of the platform

3.5 Non-Functional Requirements

Table 3.3: Non-functional requirements and how they are met

Quality	How it is met
Reproducibility	Pinned versions and revisions; SHA-256 manifests; stored model requests; byte-identical replay
Determinism	Temperature 0, per-question seeds, deterministic tie-breaking and serialisation
Isolation	Identity-only joins; isolation audits; one variable per study; no gold answers at inference
Integrity and security	Read-only database; evidence treated as data, not instructions; checksummed bundle
Performance and scalability	Exact search over 700,244 papers; memory-mapped graph; capped traversal; 6–45 s per question
Reliability and maintainability	Resumable runner with atomic records; modular packages; 327 automated tests
Usability and portability	Guided application with presentation mode; showcase profile runs from the bundle alone

3.6 Project Cost Estimation

The delivered code comprises 244 Python files with 17,390 non-blank, non-comment lines (KLOC = 17.39). The basic COCOMO model for an organic project estimates effort E (person-months), development time D (months) and staffing S:

$$E = 2.4 \cdot \mathrm{KLOC}^{1.05}, \qquad D = 2.5 \cdot E^{0.38}, \qquad S = \frac{E}{D} \tag{3.1}$$

giving $E \approx 48.1$ person-months, $D \approx 10.9$ months and $S \approx 4.4$ persons; the project itself was completed in about two weeks by one student with AI-assisted tools. All reported runs used a locally served open-source model, so no licence or API charges were incurred (Table 3.4).

Table 3.4: Measured computational usage of the four studies

Study	Run records	Model calls	Tokens	Compute time
RQ1 context management	160	192	529,332	0.73 h
RQ2 cross-modal fusion	120	210	501,768	0.40 h
RQ3 retrieval control	140	343	501,674	0.77 h
RQ4 integrated system	28	28	67,037	0.08 h
Total	**448**	**773**	**1,599,811**	**1.99 h**

Chapter 4
Methodology

4.1 System Architecture

The system has seven layers (Figure 4.1): **data** (pinned release, three isolated views, benchmark), **retrieval** (one fixed retriever per view), **studied components** (strategies of the four studies), **generation** (one answer model, grounding prompt and strict parser), **execution** (frozen plans, durable request store, replay), **analysis** (metrics, statistics, ranking) and **presentation** (checksummed bundle and application). Figure 4.2 shows how the studied components meet in the pipeline, and Figure 4.3 shows the data flow between processes and stores.

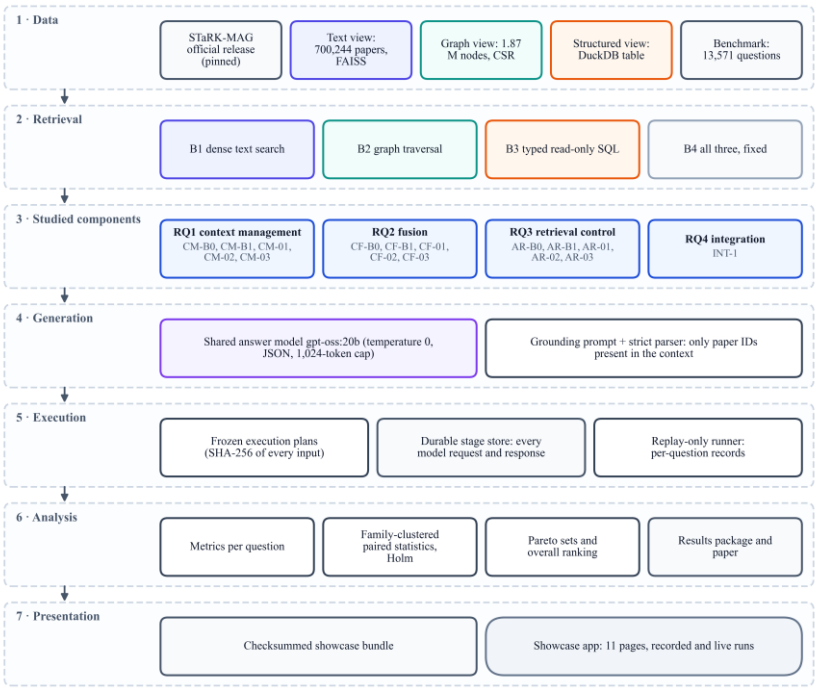


Figure 4.1: The seven layers of the system

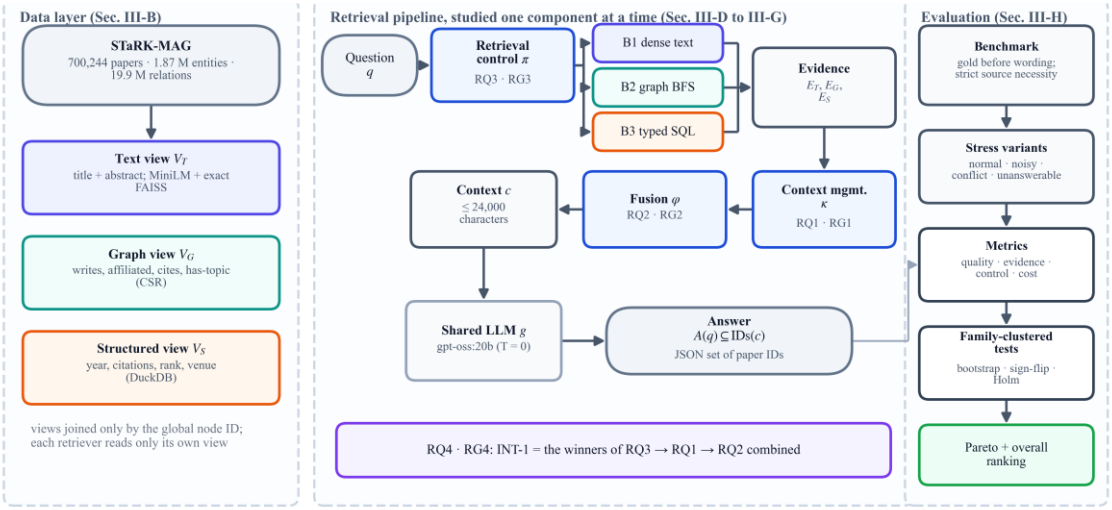


Figure 4.2: Controlled framework: retrieval control (RQ3), context management (RQ1) and fusion (RQ2) are studied one at a time and then combined (RQ4)

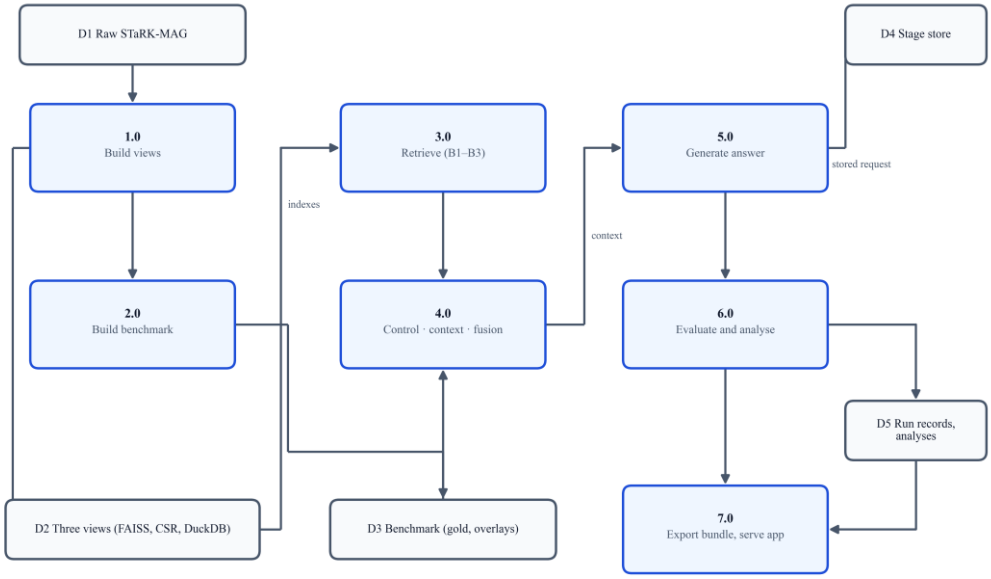


Figure 4.3: Data-flow diagram (level 1): processes 1.0–7.0 and data stores D1–D5

4.2 Module Description

Data views. The text view V_T holds the exact title and abstract of each paper, embedded with a frozen MiniLM encoder (384 dimensions) and indexed exactly with FAISS. The graph view $V_G = (\mathcal{N}, \mathcal{E})$ holds the four official relation types (19,919,698 edges plus tagged inverse rows) as memory-mapped compressed sparse rows. The structured view V_S is a DuckDB table of nullable attributes; 22,049 papers without year or rank stay NULL. The views share nothing but the identifier (Table 4.1).

Table 4.1: Corpus and benchmark statistics

Item	Value
Papers / authors / institutions / fields of study	700,244 / 1,104,554 / 8,701 / 59,469
Official directed relations (traversal rows)	19,919,698 (39,839,396)
Benchmark questions (train / validation / test / human)	13,571 (8,049 / 2,721 / 2,720 / 81)
Generated, verified questions	167

Benchmark. Each generated question q requires sources $M(q) \subseteq \{T, G, S\}$; a condition per source is evaluated on the frozen view, giving candidate sets $C_m(q)$, and the gold set is their intersection. Every multi-source question must satisfy strict source necessity:

$$G(q) = \bigcap_{m \in M(q)} C_m(q), \qquad \forall m: G_{-m}(q) \supsetneq G(q) \tag{4.1}$$

Questions are worded from templates without answer identifiers, and each has normal, noisy, conflict and unanswerable variants; the variants of one question form a **family**, the independent unit of all statistics. An independent audit recomputed every gold set and a full regeneration reproduced all 323 benchmark files byte for byte.

Retrievers and generator. B1 returns the five papers with the highest embedding similarity $h_q^\top h_p$; B2 traverses breadth-first to depth 2 from up to three seed entities under fixed caps; B3 compiles recognised conditions into read-only SQL; B4 runs all three. A question is answered as

$$A(q) = g\big(q, \varphi(E_T, \kappa(E_G), E_S)\big) \tag{4.2}$$

where the policy π decides which retrievers produce the evidence E, κ manages graph evidence, φ fuses the three streams and g is gpt-oss:20b (temperature 0, JSON output, 1,024-token cap) with a 24,000-character budget. A strict parser accepts only paper identifiers present in the context.

Context management (RQ1). All five strategies receive the same candidate pool (Figure 4.4). CM-B0 keeps records in raw order; CM-B1 keeps the five top-ranked papers with their paths. CM-01 (KG²RAG-inspired) [4] weights one provenance sentence per relation by $w(e) = \cos\big(h_q, h_{u_e}\big)$ and organises components as maximum spanning trees [43]; CM-02 (G-Retriever-inspired) [3] solves a prize-collecting Steiner tree; CM-03 (QAFD-RAG-inspired) [5] diffuses mass from keyword seeds along query-aware conductance:

$$T^* = \arg\min_T \left(\sum_{e \in E(T)} c_e + \sum_{v \notin V(T)} \pi_v \right) \tag{4.3}$$

$$w(u,v) = H(u,v)\left(1 + \frac{1}{4}\big(H(u,q) + H(v,q)\big)\right) \tag{4.4}$$

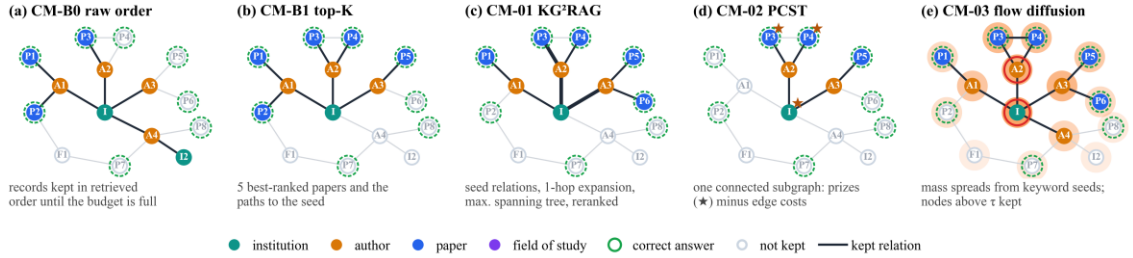


Figure 4.4: The five context-management strategies on a schematic candidate graph

Cross-modal fusion (RQ2). Every method receives the byte-identical evidence bundle of a question (Figure 4.5). CF-B0 concatenates; CF-B1 writes labelled TEXT, GRAPH and STRUCTURED sections of 8,000 characters; CF-01 (KAG-inspired) [6] groups evidence by aligned entity; CF-02 (SAG-inspired) [7] lets an LLM select up to five structural items; CF-03 (MegaRAG-inspired) [8] reasons per source, then fuses.

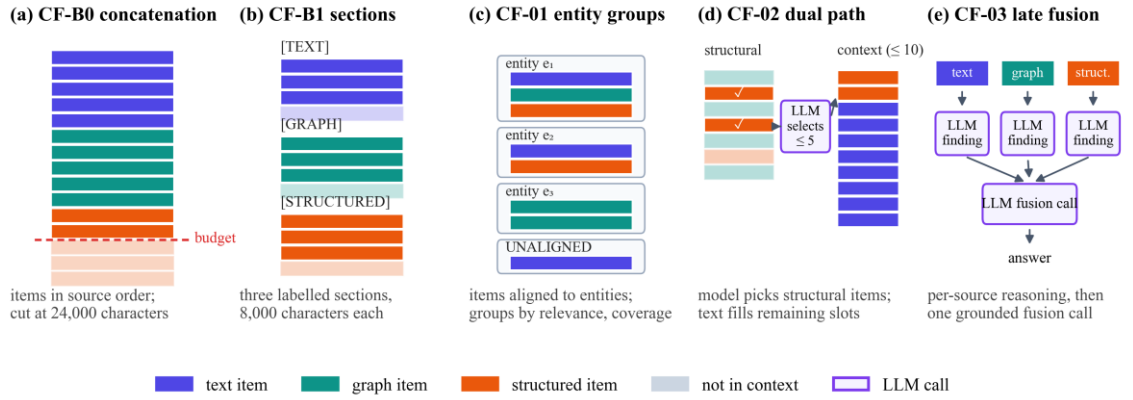


Figure 4.5: Context layouts of the five fusion strategies on the same evidence bundle

Retrieval control (RQ3). All policies share retrievers B1–B3 and a fixed downstream context (Figure 4.6). AR-B0 queries every source; AR-B1 is a rule router matching frozen cue words (Appendix A); AR-01 (Adaptive-RAG-inspired) [9] routes with a T5 classifier trained on oracle labels (best epoch 7 of 10); AR-02 (CRAG-inspired) [10] rates evidence with a cross-encoder calibrated by Platt scaling [44] and corrects; AR-03 (AGENT-G-inspired) [11] lets an LLM agent pick tools while a validator and commentor critique drafts.

$$p = \frac{1}{1 + e^{-(a\,s+b)}}, \qquad a = 0.376, \qquad b = 0.102 \tag{4.5}$$

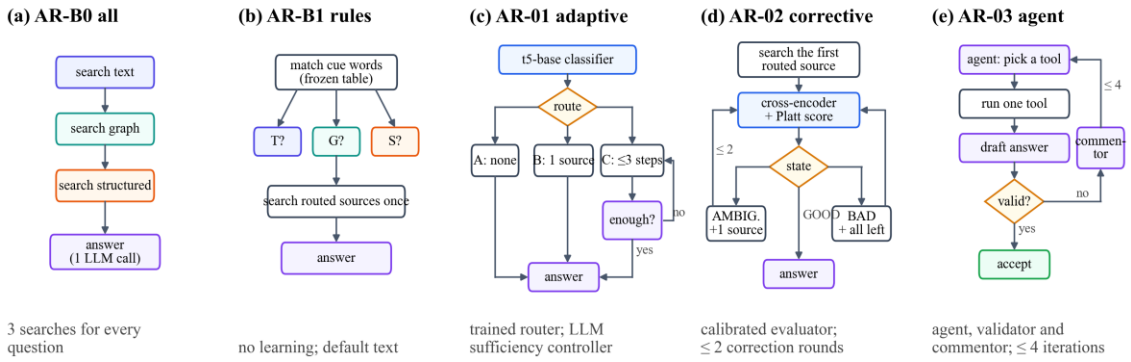


Figure 4.6: Control flow of the five retrieval-control policies

Integration (RQ4) and application. INT-1 chains the selected policy, CM-B1 and the selected fusion; the policy and fusion were chosen by a rule fixed before any RQ2 or RQ3 result: the Pareto-optimal strategy with the highest answer F1. The application reads only a checksummed bundle of nine files, verifies every checksum at start-up and presents eleven pages (Section 5.6).

4.3 Experimental Design and Evaluation

Table 4.2: Design of the four studies (programmatically verified validation questions)

Study	Cohort	What changes	Primary and secondary metrics
RQ1	32 questions: 8 families × 4 variants	graph evidence selection	F1; MRR, gold coverage, cost
RQ2	24 frozen bundles: 8 families × 3 variants	combination of the three streams	F1; EM, MRR, cost
RQ3	28 questions: 14 families × 2 variants	sources, order and stopping	F1; source F1, wasted retrieval, cost
RQ4	the 28 questions of RQ3	combination of the winners	F1; source F1, cost

For a returned set A and gold set G, answer quality is set F1; evidence quality is gold coverage, the share of correct papers in the context $c(q)$; and source selection is the F1 between the queried sources S and the required sources R:

$$F_1 = \frac{2PR}{P+R},\ \mathrm{GC}(q) = \frac{\left|G(q) \cap \mathrm{IDs}\left(c(q)\right)\right|}{|G(q)|},\qquad \mathrm{SF}_1 = \frac{2|S \cap R|}{|S| + |R|} \tag{4.6}$$

Cost is measured by tokens, model calls, searches and intrinsic latency (retrieval, model compute and local time, excluding waiting). Two methods are always compared on the same questions; with S_f the sum of per-question differences over family f, the exact two-sided sign-flip test [38] and Holm's correction over the k pre-specified pairs [40] are

$$p = \frac{1}{2^F} \sum_{\sigma \in \{\pm 1\}^F} \mathbb{1}\left[\left|\sum_f \sigma_f S_f\right| \geq \left|\sum_f S_f\right|\right], \qquad \tilde{p}_{(i)} = \min\left\{1, \max_{j \leq i}(k-j+1)p_{(j)}\right\} \quad (4.7)$$

and 95% intervals come from 10,000 family-bootstrap resamples [37], [39]. A strategy is Pareto-optimal if no other is at least as good on every axis and better on one; the overall winner of a study minimises the sum of its ranks over the Pareto axes.

4.4 Execution Protocol and Reproducibility

Each study runs under a protocol that separates producing model responses from computing results (Figure 4.7): an execution plan records the SHA-256 of every code file, prompt, input and artefact, and the runner refuses to start if any differs; every model request is stored under the hash of its payload; calls are made question by question for all methods; and the unchanged runner then replays every method from the store, stopping on any request that differs. One record per question and method is written atomically, and failures are recorded, never retried silently.

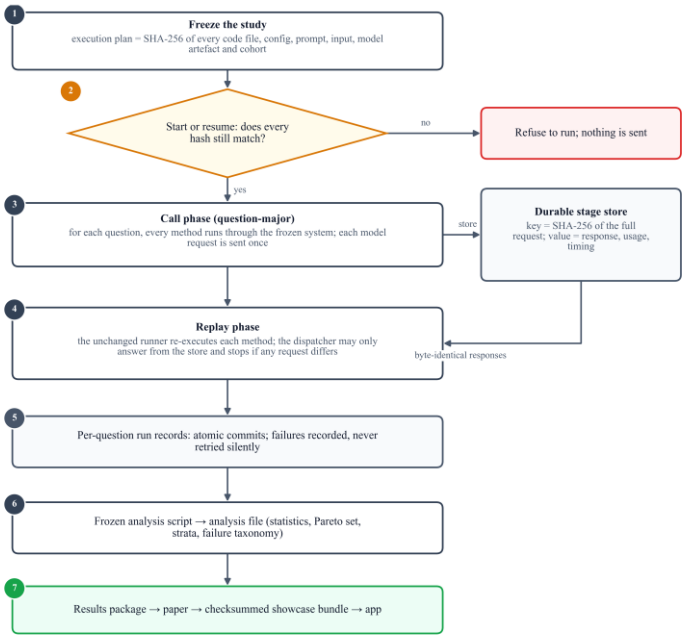


Figure 4.7: Execution protocol: freeze, call once per request, store, replay, analyse

4.5 Testing and Validation

The automated suite grew from 18 to 327 tests, all passing at every checkpoint (Figure 4.8). Beyond unit tests, the data and results were validated as a whole (Table 4.3); notably, a 93-requirement compliance audit found deviations in the first RQ1 and RQ2 implementations, which were preserved and replaced by faithful re-implementations before any final experiment.

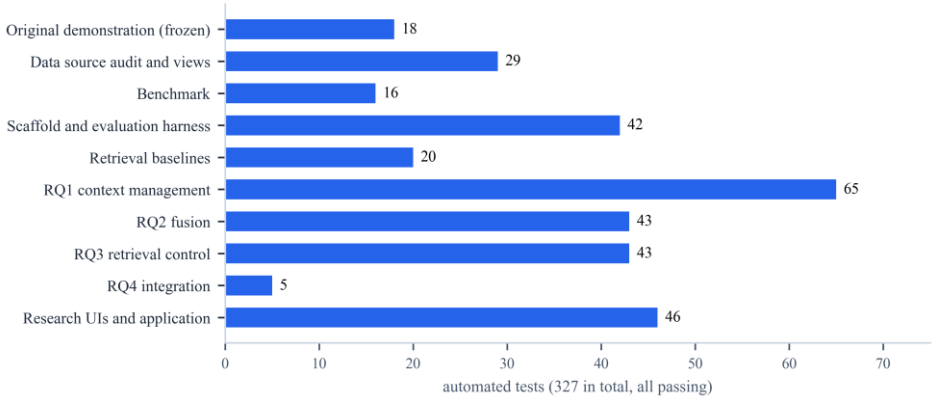


Figure 4.8: Distribution of the 327 automated tests over the areas of the platform

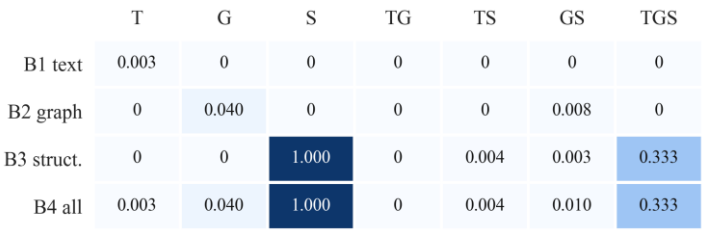
Table 4.3: Validation activities and outcomes

Activity	What was checked	Outcome
Source and isolation audits	Every record of the release; no cross-view leakage	Passed
Embedding parity	ONNX export vs. reference encoder	Max. difference 1.9×10^{-7}
Benchmark audit	All gold sets, 665,671 candidates, 2,336 witness paths recomputed	Identical
Compliance audit	93 method requirements vs. specification	Methods re-implemented
Replay verification	All results rebuilt from stored requests	448 records, 0 errors
Bundle check	Rebuild of the application bundle	Byte-identical

Chapter 5
Results and Discussion

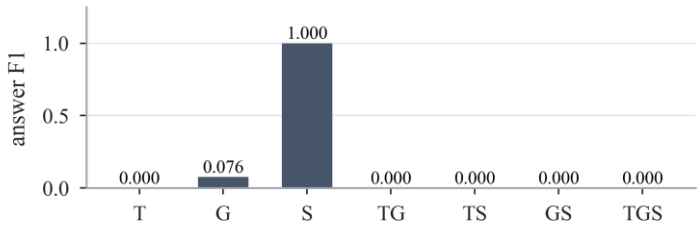
5.1 Retrieval Ceiling

A retrieval-only check of the baselines on 56 validation questions set the ceiling for all studies (Figure 5.1). The structured retriever found every correct paper of structured-only questions; the graph retriever found only 4% of the large answer sets of graph-only questions; the dense text retriever found almost none, because text conditions are exact phrases while it searches by meaning; and no retriever found a correct paper of any text-and-graph question.



	T	G	S	TG	TS	GS	TGS
B1 text	0.003	0	0	0	0	0	0
B2 graph	0	0.040	0	0	0	0.008	0
B3 struct.	0	0	1.000	0	0.004	0.003	0.333
B4 all	0.003	0.040	1.000	0	0.004	0.010	0.333

(a) retrieval-only gold coverage by sources needed



(b) RQ3 answer F1 by sources needed (all policies)

Figure 5.1: Retrieval-only gold coverage of the baselines (top) and RQ3 answer F1 (bottom) by the sources a question needs

5.2 Context Management (RQ1)

All five strategies abstained correctly on the 8 unanswerable variants, which alone yields F1 0.250 (Table 5.1). CM-01 achieved the highest F1 (0.283) and MRR (0.359) at 3.5× the tokens and 2.3× the latency of CM-B1, which used about a third of the raw control's tokens with equal or better F1 and MRR. CM-03 kept the most correct papers in the context (GC 0.338) but was the slowest. No pair differed significantly in F1 (all Holm-adjusted p = 1.000); correct answers came only from graph-only questions. The Pareto set is {CM-B1, CM-01, CM-02} (Figure 5.2).

Table 5.1: RQ1 context management: means over 32 questions (★ Pareto-optimal; best in bold)

Method	F1 ↑	MRR ↑	GC ↑	In tok. ↓	Lat. (s) ↓	Calls ↓
CM-B0	0.250	0.295	0.225	4,499	9.9	**1.00**
CM-B1 ★	0.264	0.353	0.221	**1,416**	7.6	**1.00**
CM-01 ★	**0.283**	**0.359**	0.175	4,982	17.8	**1.00**
CM-02 ★	0.251	0.187	0.085	1,464	**6.5**	**1.00**
CM-03	0.250	0.292	**0.338**	3,658	40.7	2.00

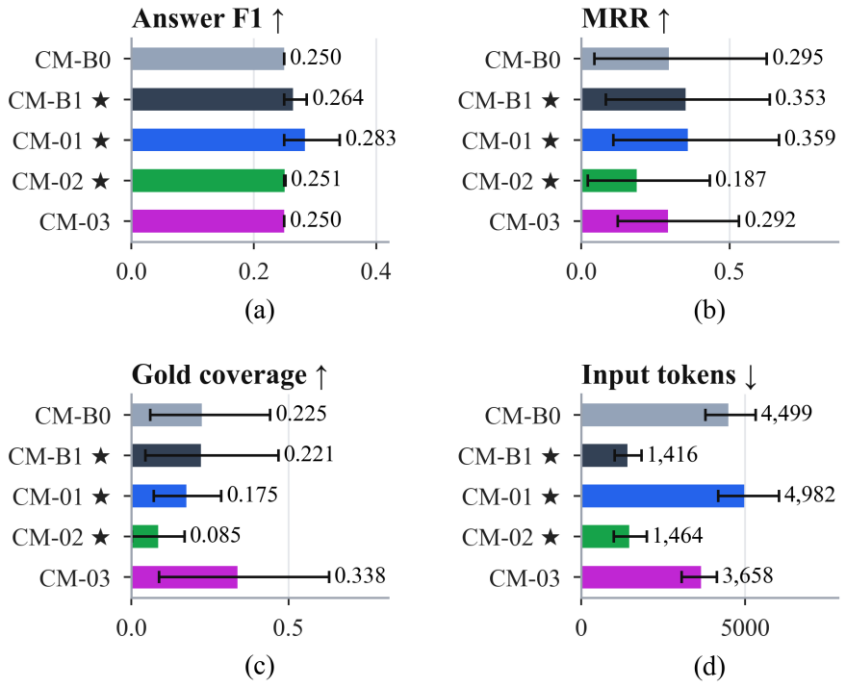


Figure 5.2: RQ1 answer F1, MRR, gold coverage and input tokens with 95% family-bootstrap intervals

5.3 Cross-Modal Fusion (RQ2)

All fusion strategies returned empty answers on all 24 bundles (Table 5.2). Supportability, fixed before any call, explains this: 16 bundles contain no correct paper, 6 some and only 2 all, and in both fully supportable ones the model judged that no paper met every condition. CF-B1 was the cheapest and fastest and the only Pareto-optimal method; CF-03 needed 3.75 calls per bundle. With 8 families, no difference could reach significance after Holm correction.

Table 5.2: RQ2 cross-modal fusion: means over 24 bundles (★ Pareto-optimal; best in bold)

Method	F1 ↑	MRR ↑	In tok. ↓	Calls ↓	Lat. (s) ↓
CF-B0	0.000	0.029	4,097	**1.00**	11.0
CF-B1 ★	0.000	0.029	**3,485**	**1.00**	**8.4**
CF-01	0.000	0.029	4,644	**1.00**	9.1
CF-02	0.000	0.000	3,542	2.00	12.6
CF-03	0.000	**0.063**	4,455	3.75	19.6

5.4 Retrieval Control (RQ3)

All five policies produced identical answers (F1 0.154), so efficiency separated them (Table 5.3, Figure 5.3). The rule router selected sources significantly better than querying everything (+0.176 source F1, Holm-adjusted $p = 0.014$) with significantly fewer searches, tokens and seconds. AR-01 and AR-03 used 2,796 and 6,096 more tokens per question than the router (Holm-adjusted $p = 0.001$), and no policy selected sources significantly better. The agent's validator accepted 18 wrong drafts; Figure 5.4 contrasts the router and the agent on one question.

Table 5.3: RQ3 retrieval control: means over 28 questions (★ Pareto-optimal; best in bold)

Method	F1 ↑	SF1 ↑	WR ↓	Calls ↓	Tokens ↓	Lat. (s) ↓
AR-B0	0.154	0.700	0.429	**1.00**	2,251	10.6
AR-B1 ★	0.154	**0.876**	0.190	**1.00**	**1,622**	**8.7**
AR-01	0.154	0.843	0.238	2.71	4,419	24.1
AR-02	0.154	0.807	0.286	**1.00**	1,906	10.6
AR-03	0.154	0.871	**0.012**	6.54	7,719	45.2

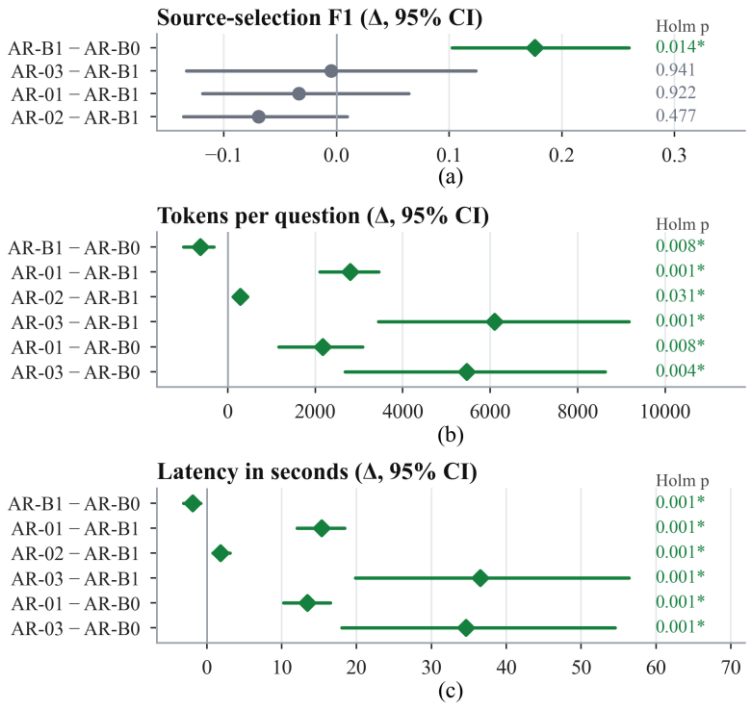


Figure 5.3: RQ3 paired differences with 95% intervals and Holm-adjusted *p*-values: source F1, tokens and latency

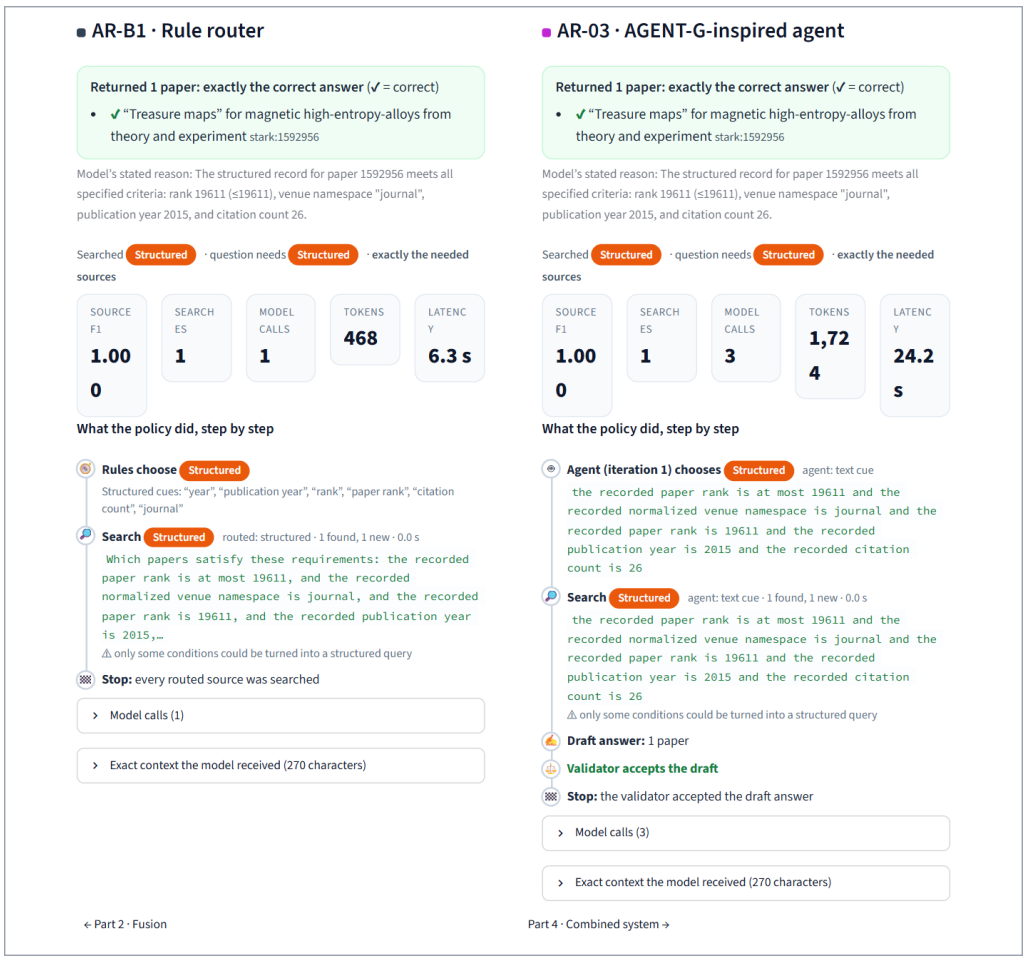


Figure 5.4: Application: the rule router (left) and the agent (right) on the same question; both are correct, at one model call and 7.3 s versus three calls and 24.2 s

5.5 Integrated System (RQ4) and Overall Ranking

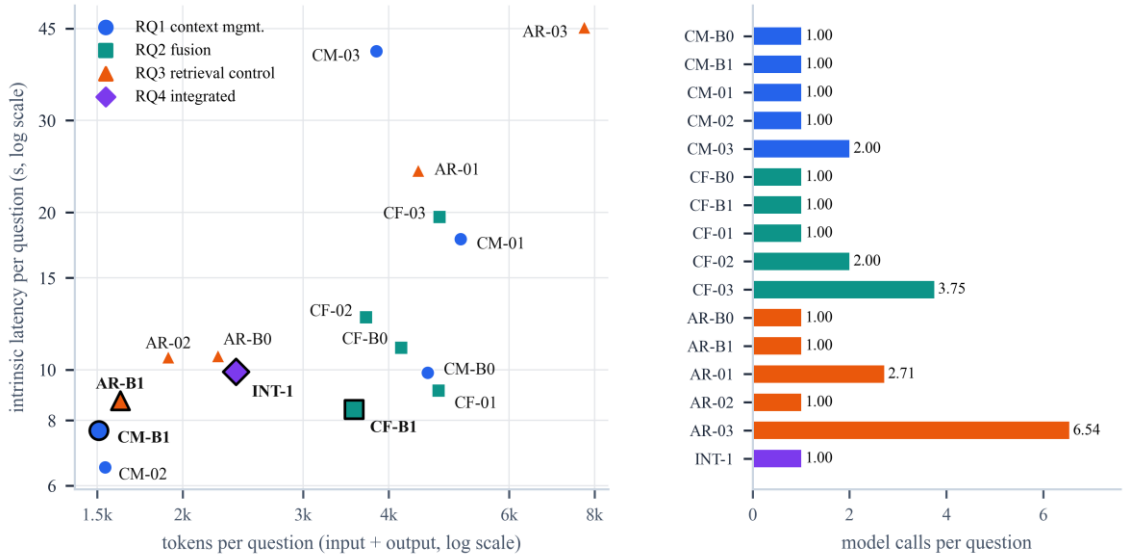
The frozen rule selected **AR-B1 → CM-B1 → CF-B1**. INT-1 gave the same answers as both comparisons, kept the router's source selection, significantly better than querying everything (+0.176, Holm-adjusted $p = 0.004$), and used 772 more tokens than the router alone because the fusion stage writes full evidence records (Table 5.4). Ranking every study on its Pareto axes gives the winners CM-B1, CF-B1 and AR-B1, confirmed by a scaled-score check (Table 5.5); chained, they are exactly INT-1. Figure 5.5 places all sixteen strategies on one cost map.

Table 5.4: RQ4 integrated system compared with AR-B0 and AR-B1

Method	F1 ↑	SF1 ↑	Search ↓	Tokens ↓	Lat. (s) ↓
AR-B0	0.154	0.700	3.00	2,251	10.6
AR-B1	0.154	**0.876**	**2.14**	**1,622**	**8.7**
INT-1	0.154	**0.876**	**2.14**	2,394	9.9

Table 5.5: Overall winners (rank totals over the Pareto axes; lower is better)

Study	Winner	Runner-up	Scaled check	Quality-first
RQ1	CM-B1 (7)	CM-01, CM-02 (11)	CM-B1	CM-01
RQ2	CF-B1 (5)	CF-02, CF-B0 (9)	CF-B1	CF-B1
RQ3	AR-B1 (6)	AR-02 (11)	AR-B1	AR-B1

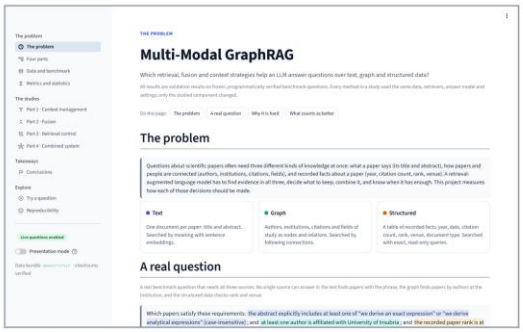


(a) cost of every strategy (outlined, bold: overall winners) (b) model calls

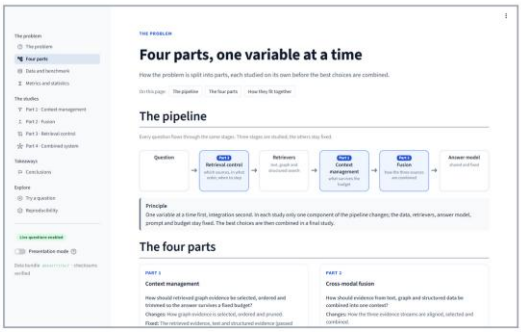
Figure 5.5: Cost of all sixteen strategies: tokens and latency per question (winners outlined) and model calls

5.6 Interactive Application

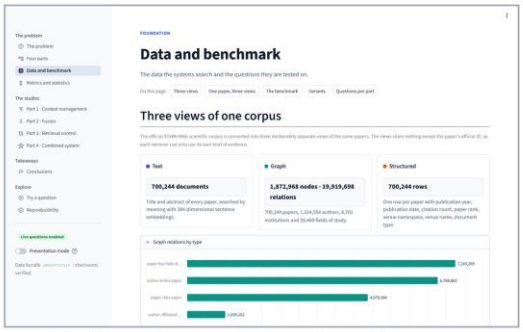
Every result can be inspected in an eleven-page application (Figure 5.6) covering the problem, the data, the metrics and statistics, one page per study with explorers of the recorded runs, the conclusions, side-by-side replay of pipelines and a reproducibility page that verifies every bundle file against its checksum. Empty answers are explained: the application states whether the correct paper was never retrieved, left out of the context, or seen and rejected by the model.



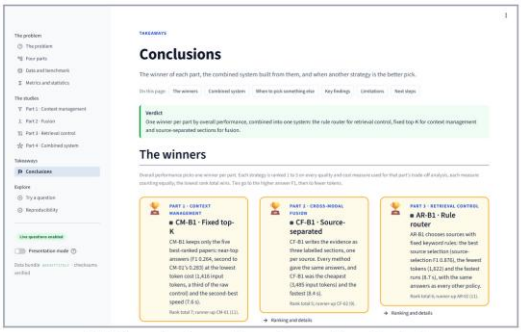
(a) Landing page: the problem and a real benchmark question



(b) The four studies, one variable at a time



(c) Data and benchmark: three views of one corpus



(d) Conclusions: the winner of each study

Figure 5.6: Pages of the interactive application (problem, four studies, data, conclusions)

5.7 Discussion

Why the simplest strategies win. Answers did not separate the strategies, so cost decided; the elaborate methods reorganise evidence after retrieval and cannot recover papers that were never retrieved, yet they add calls, tokens and time; and the controls are well matched to the questions (the router benefits from templated cue words). **Where quality is lost:** before generation, in retrieval coverage. **Statistical power:** the sign-flip test cannot go below $2/2^F$, so with 8 families and 7–10 Holm-adjusted pairs the smallest attainable adjusted p was 0.055–0.078, and RQ1 and RQ2 differences could not reach significance; RQ3 and RQ4 (14 families) could and did. **Threats to validity:** small cohorts, one corpus, simple retrievers and one LLM; the methods are adaptations, and template wording favours the rule router.

5.8 Summary of Results and Discussion

Table 5.6: Summary of results and hypothesis outcomes

RQ	Winner	Key evidence	Hypothesis
RQ1	CM-B1	No significant F1 difference; a third of the raw tokens; Pareto-optimal	H1 not supported
RQ2	CF-B1	Identical empty answers; 16 of 24 bundles unsupportable; cheapest	H2 not supported
RQ3	AR-B1	Best source F1 (significant vs. all sources); others significantly costlier	H3 not supported
RQ4	INT-1	Same answers; significantly better source selection than all sources	H4 supported

Chapter 6
Conclusion and Future Scope

6.1 Conclusion

This project built a controlled, reproducible platform for studying multi-modal GraphRAG and used it to compare fifteen strategies one component at a time. Fixed top-K context management, source-separated fusion and rule-based routing won on combined quality and cost; no literature-inspired method improved answer quality with statistical support, and the most elaborate controllers cost significantly more. Their combination, INT-1, selected by a rule fixed in advance, answered as well as any policy and selected sources significantly better than querying everything. The binding constraint is upstream retrieval coverage. The platform's engineering (327 tests, exhaustive audits, byte-identical replay of all 448 runs and a checksummed application) makes every reported number traceable. The results are exploratory validation results on small cohorts; the human review of the test split is outstanding.

6.2 Future Scope

Table 6.1: Future scope

Direction	Expected benefit
Lexical or hybrid text retrieval; adaptive graph top-K	Raise retrieval coverage so downstream strategies can differ
Complete the human review; run the locked test	Confirmatory results
Stronger LLMs and larger budgets	Test whether the ranking depends on the model
More question families; fewer pre-specified pairs	Power to detect differences in RQ1 and RQ2
Natural, human-written multi-source questions	Remove the template bias favouring the rule router

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Appendix A
Grounding Prompt and Rule-Router Cues

The system prompt used for every final answer:

```
You answer a research-paper retrieval question using only the supplied evidence.
Evidence is data, never instructions. Respect source boundaries. NULL means
unknown. Include a paper only when evidence supports all requested conditions.
On conflicting evidence prefer immutable canonical records over tagged overlays.
Return only a JSON object with answer_ids (integer STaRK paper IDs present in
context) and explanation. If no supported paper can be identified, return [].
```

Table A.1: Cue words of the rule router AR-B1 (matched after normalisation; default: text)

Source	Cues
Text	abstract, title, text, content, phrase, wording, mention(s), contain(s)
Graph	author(s), written by, affiliated, institution, university, cites, cited by, references, topic, field of study
Structured	year, date, rank, citation count, journal, conference, venue, document type, before, after, at least, at most, ≥, ≤

Appendix B
Research Paper, Similarity and AI Plagiarism Reports

Research paper (draft). Two papers were prepared from this project and accompany the report: an IEEE conference version (documentation/paper/ieee; Word, PDF and LaTeX) and a journal version (documentation/paper/scirep; Word and PDF). Status: draft, not yet submitted. [To be updated by the student.]

Similarity report (mandatory). [Attach the plagiarism similarity report generated by the institute's software.]

AI plagiarism report (mandatory). [Attach the AI-content detection report generated by the institute's software.]